ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

Independent V3 design: 360–440 V DC input; hazardous non-isolated 200 V output.
Off-board choke, contactors and current sensor require the system assembly drawing.

Protected electronics supply and DSP power sequencing

AUX_OK_N
XRS_N

文件: 01_supply.kicad_sch

TMS320F28335 controller and signal assignment

ARMED
AUX_OK_N
CURRENT_OK
VOLTAGE_OK
ADCREFM
ADCREFP
ADCRESEXT
ADC_BUS
ADC_CURRENT
ADC_IREF
ADC_OUT
ADC_RAW
ADC_TEMP
ADC_V12
ARM_PULSE
BIAS_EN
BOOT84
BOOT85
BOOT86
BOOT87
CLK30
DRV_RESET
EMU0
EMU1
EPWM_A
EPWM_B
FAN_TACH
FLT_H_N
FLT_L_N
MAIN_CMD
MAIN_FB
PG_H_N
PG_L_N
PG_S_N
PRE_CMD
PRE_FB
RDY_H
RDY_L
RUN_ENABLE
TCK
TDI
TDO
THERM_OK
TH_H_PWM
TH_L_PWM
TMS
TRST
UART_RX
UART_TX
XRS_N

文件: 02_dsp.kicad_sch

DSP per-pin bypass and ADC reference

ADCREFM
ADCREFP
ADCRESEXT

文件: 03_bypass.kicad_sch

Clock, boot, JTAG and communications

TDO
BOOT84
BOOT85
BOOT86
BOOT87
CLK30
EMU0
EMU1
TCK
TDI
TMS
TRST
UART_RX
UART_TX

文件: 04_debug.kicad_sch

Isolated voltage receiver and bipolar current conditioning

ADC_BUS
ADC_CURRENT
ADC_IREF
ADC_OUT
ADC_RAW
ADC_TEMP
ADC_V12
BUS_N
BUS_P
BUS_V
I_OCD_N
I_V
OUT_N
OUT_P
OUT_V
RAW_N
RAW_P
RAW_V

文件: 05_analog.kicad_sch

Asynchronous protection, fault latch and PWM gating

AUX_OK_N
BUS_V
I_V
OUT_V
RAW_V
XRS_N
ARMED
ARM_PULSE
BIAS_EN
COIL_INHIBIT
CURRENT_OK
DRV_RESET
EPWM_A
EPWM_B
FLT_H_N
FLT_L_N
I_OCD_N
PG_H_N
PG_L_N
PG_S_N
PWM_H
PWM_L
RDY_H
RDY_L
RUN_ENABLE
THERM_OK
VOLTAGE_OK

文件: 06_trip.kicad_sch

Contactor, feedback, fan and thermal interlock

COIL_INHIBIT
FAN_TACH
MAIN_CMD
MAIN_FB
PRE_CMD
PRE_FB
THERM_OK

文件: 07_actuate.kicad_sch

Power-board harness and mechanical support

BIAS_EN
BUS_N
BUS_P
DRV_RESET
FLT_H_N
FLT_L_N
OUT_N
OUT_P
PG_H_N
PG_L_N
PG_S_N
PWM_H
PWM_L
RAW_N
RAW_P
RDY_H
RDY_L
TH_H_PWM
TH_L_PWM

文件: 08_interface.kicad_sch

ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

Sheet: /
File: sic-v3-control.kicad_sch

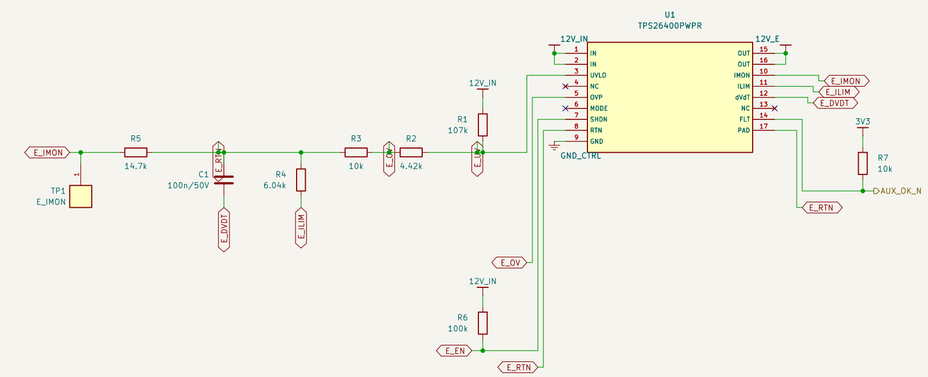
Title: V3 400V / 200V / 5kW — control

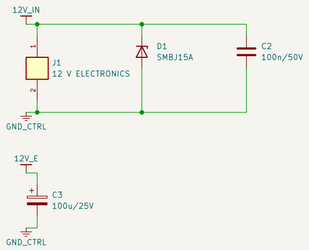
Size: A2 | Date: | **Rev: V3**
KiCad E.D.A. 10.0.7 | Id: 1/9

Protected electronics supply and DSP power sequencing

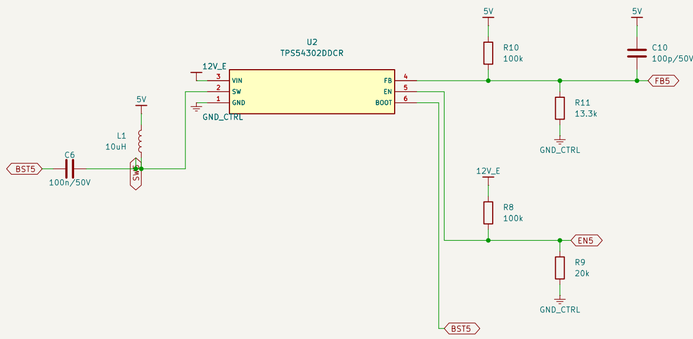
PWR_FLAG symbols state supply provenance for ERC; they do not prove voltage, current capacity or protection.

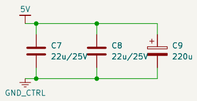
12 V electronics entry: independent 2 A protection

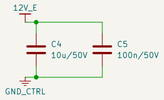




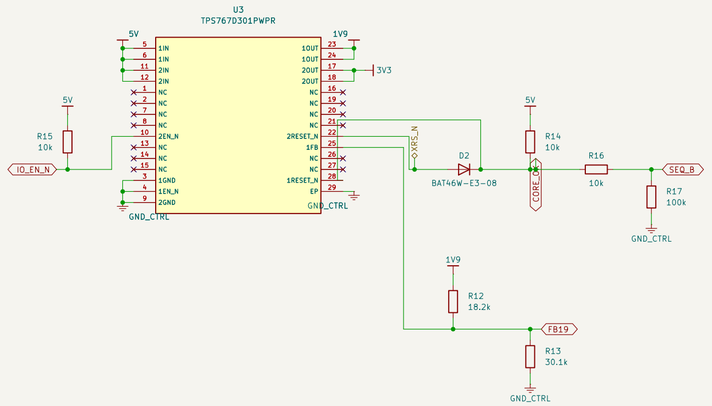
5 V logic / sensor buck supply

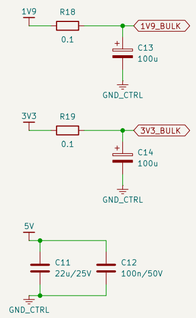




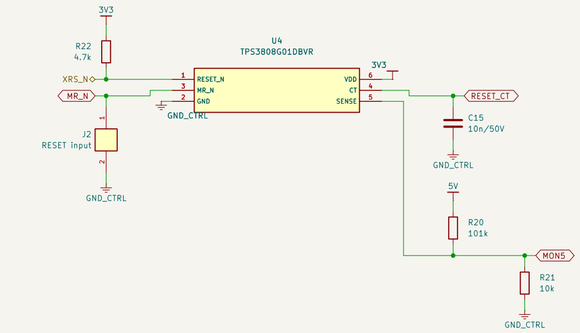


Core–first 1.9 V / 3.3 V supply sequencing





Supply supervisor; reset also clears the hardware latch





Supply provenance (PWR_FLAG)



ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

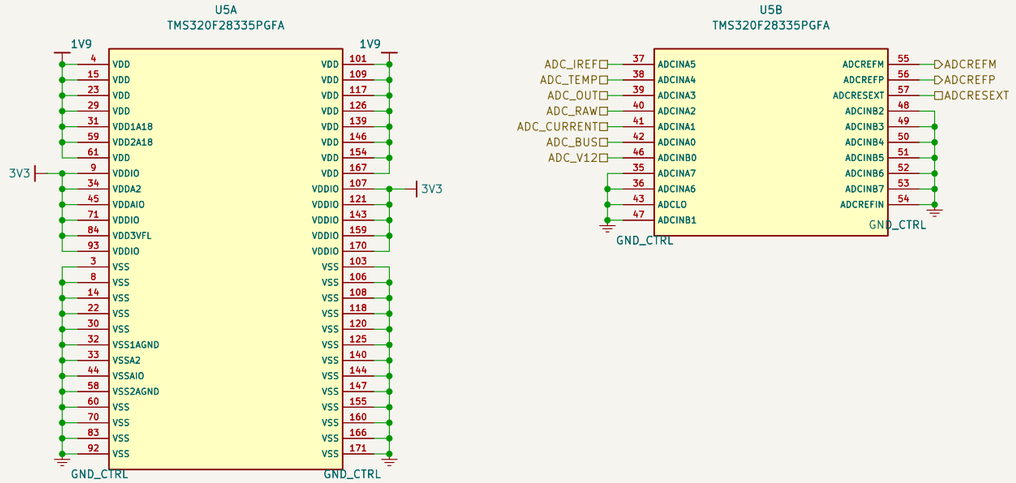
Sheet: /Protected electronics supply and DSP power sequencing/
File: 01_supply.kicad_sch

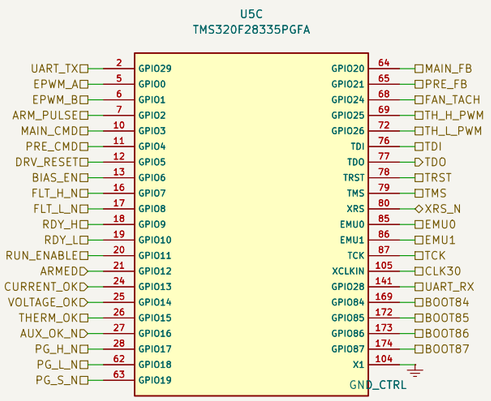
Title: Protected electronics supply and DSP power sequencing

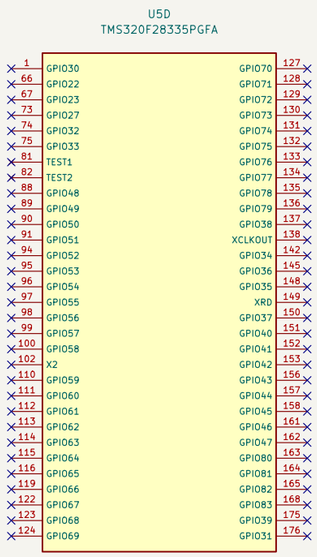
Size: A2 | Date: | Rev: V3
KiCad E.D.A. 10.0.7 | Id: 2/9

TMS320F28335 controller and signal assignment

TMS320F28335; ePWM dead-band + asynchronous TZ inputs







ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

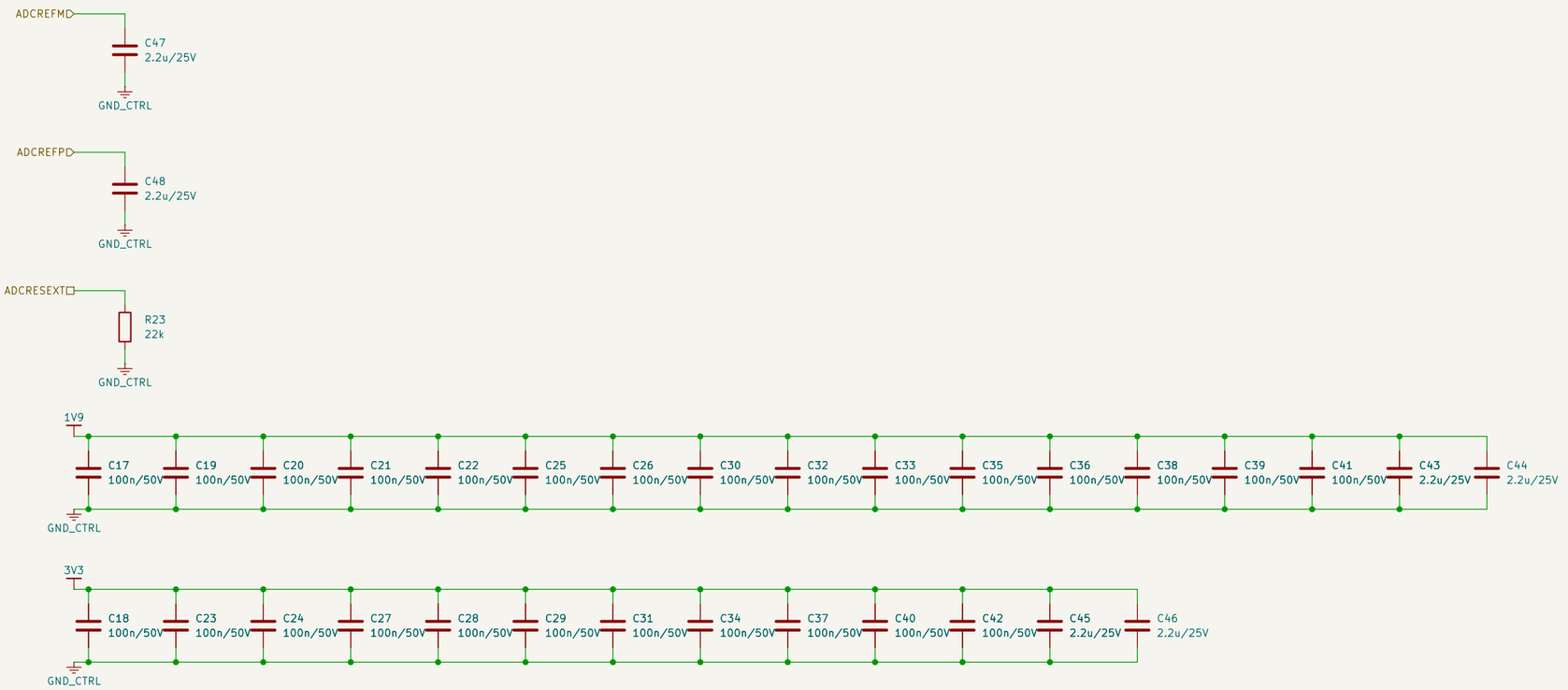
Sheet: /TMS320F28335 controller and signal assignment/
File: 02_dsp.kicad_sch

Title: TMS320F28335 controller and signal assignment

Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 3/9

DSP per-pin bypass and ADC reference

Per-pin capacitors; placement tied to actual DSP supply-pad positions



ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

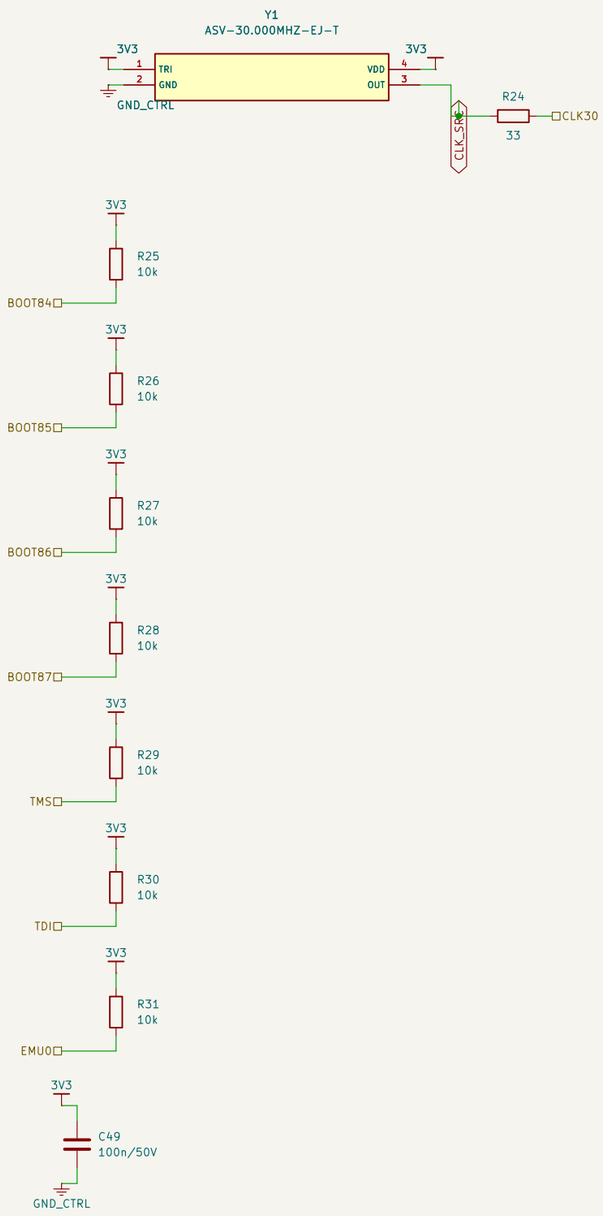
Sheet: /DSP per-pin bypass and ADC reference/
File: 03_bypass.kicad_sch

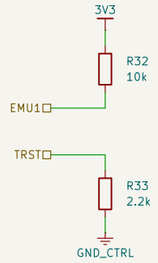
Title: DSP per-pin bypass and ADC reference

Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 4/9

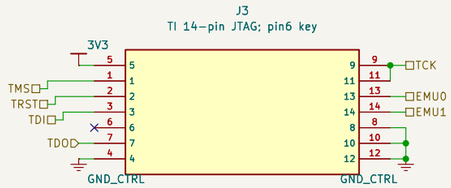
Clock, boot, JTAG and communications

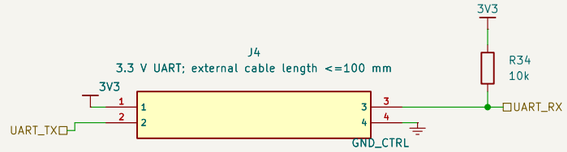
30 MHz clock and boot straps





JTAG and 3.3 V UART; accessible control domain only





ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

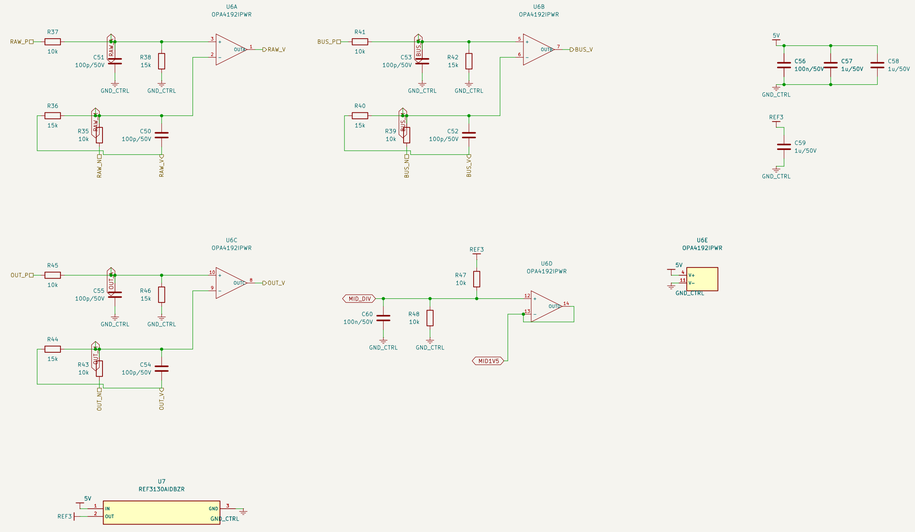
Sheet: /Clock, boot, JTAG and communications/
File: 04_debug.kicad_sch

Title: Clock, boot, JTAG and communications

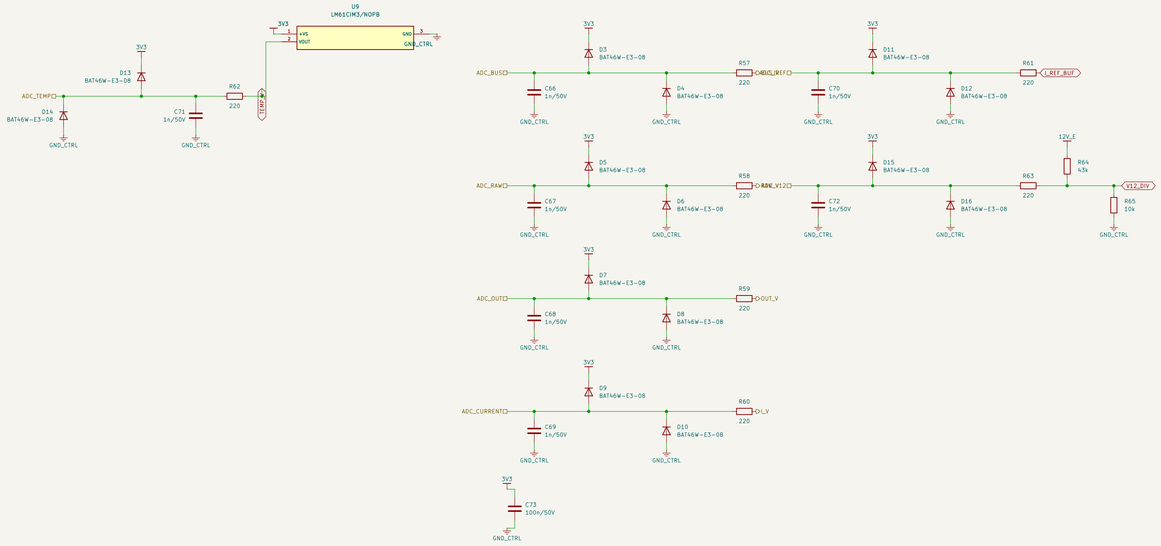
Size: A3	Date:	Rev: V3
KiCad E.D.A. 10.0.7		Id: 5/9

Isolated voltage receiver and bipolar current conditioning

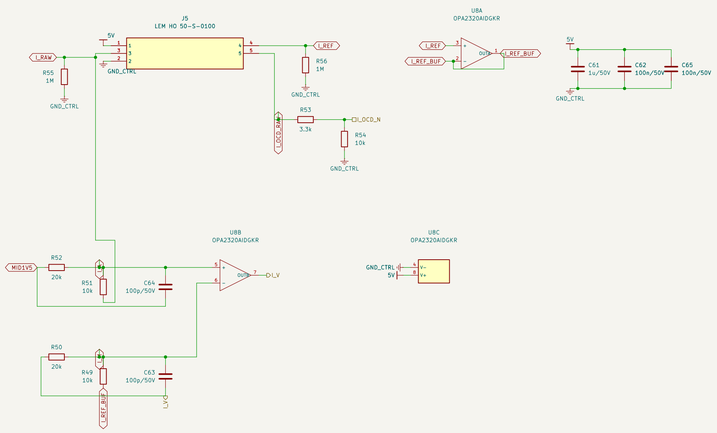
Balanced differential voltage receivers; output = 1.5 x differential input



ADC charge reservoirs and input protection



Bipolar current measurement; buffered actual sensor reference



ENGINEERING PROTOTYPE REVIEW – RELEASE BLOCKED – NOT HARDWARE QUALIFIED

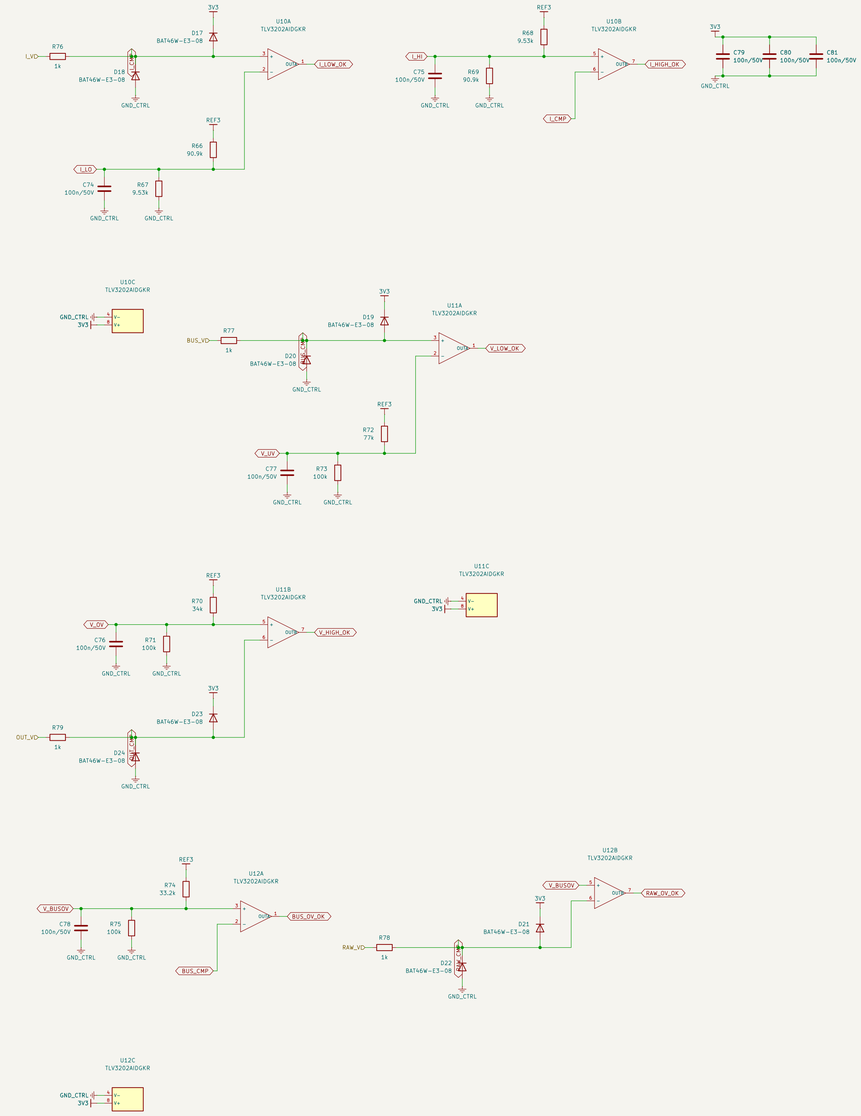
Sheet: /isolated voltage receiver and bipolar current conditioning/
File: 05_analog.kicad_sch

Title: Isolated voltage receiver and bipolar current conditioning

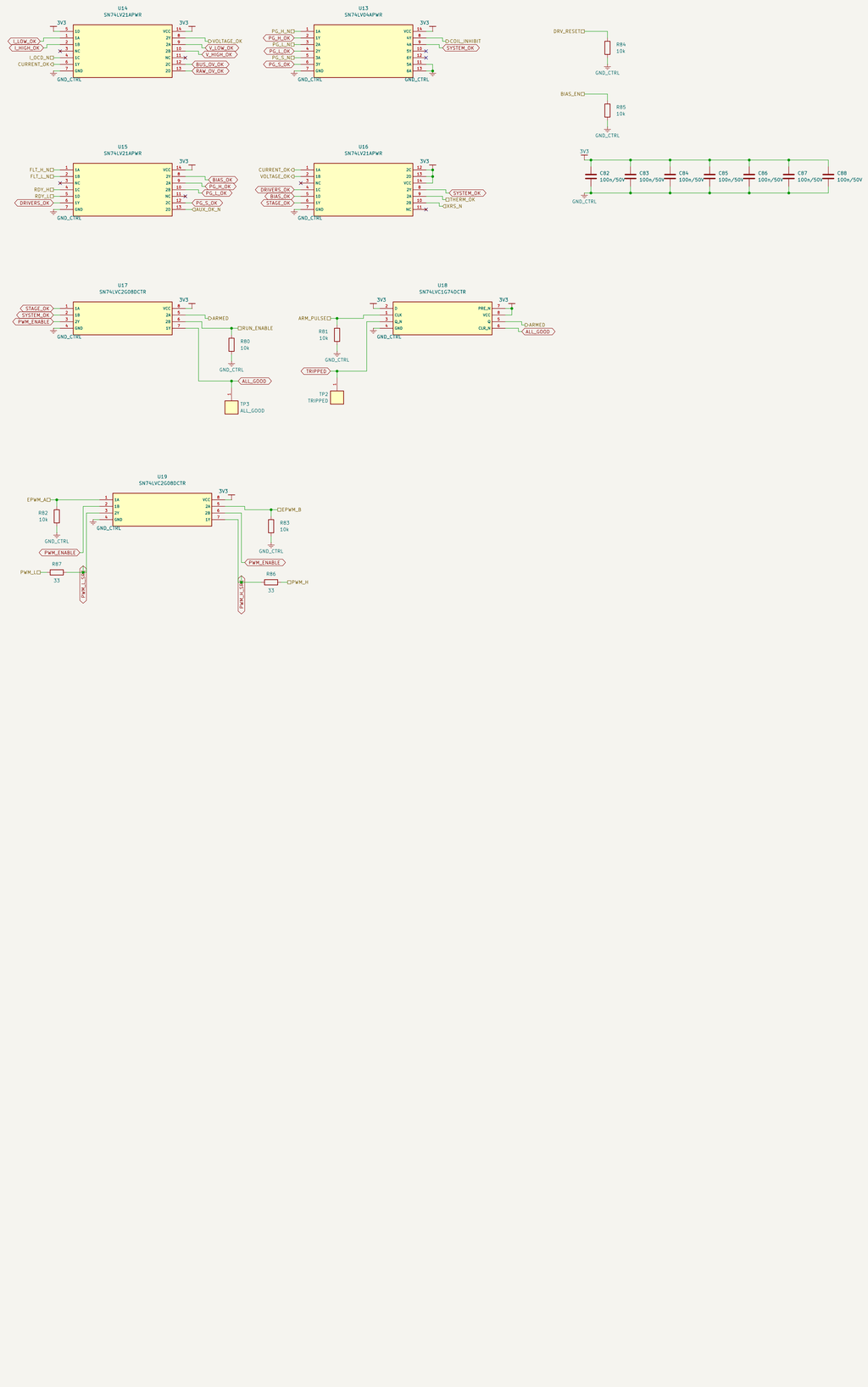
Size: A1 | Date: | Rev: V3
KiCad E.D.A. 10.0.7 | Id: 6/9

Asynchronous protection, fault latch and PWM gating

Hardware current window ±38 A, OVP225 V and UVLO340 V



Power-good polarity, asynchronous fault tree and latched inhibit



ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

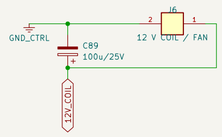
Sheet: /Asynchronous protection, fault latch and PWM gating/
File: 06_trip.kicad_sch

Title: Asynchronous protection, fault latch and PWM gating

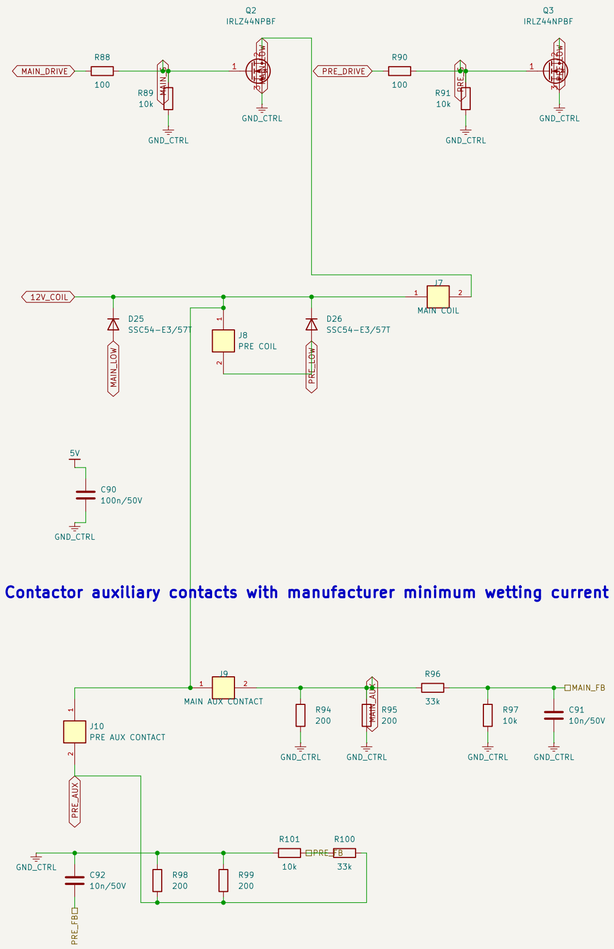
Size: A1 | Date: | Rev: V3
KiCad E.D.A. 10.0.7 | Id: 7/9

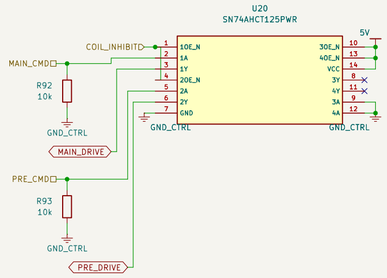
Contactor, feedback, fan and thermal interlock

Dedicated 12 V coil/fan branch; external 8 A fuse and return



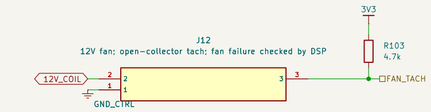
Main and precharge drivers; fail-safe default off

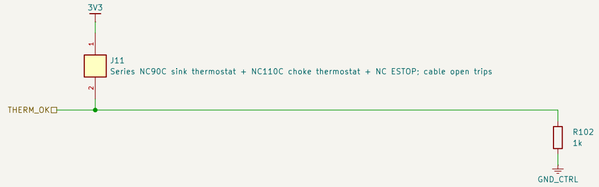




Contactor auxiliary contacts with manufacturer minimum wetting current

Fail-open thermal / emergency-stop loop and fan tachometer





ENGINEERING PROTOTYPE REVIEW — RELEASE BLOCKED — NOT HARDWARE QUALIFIED

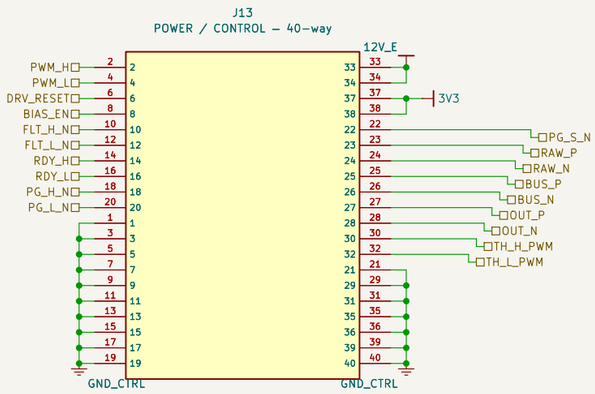
Sheet: /Contactor, feedback, fan and thermal interlock/
File: 07_actuate.kicad_sch

Title: Contactor, feedback, fan and thermal interlock

Size: A2	Date:	Rev: V3
KiCad E.D.A. 10.0.7		Id: 8/9

Power–board harness and mechanical support

Keyed SELV harness to the power board



Mechanical



ENGINEERING PROTOTYPE REVIEW – RELEASE BLOCKED – NOT HARDWARE QUALIFIED

Sheet: /Power–board harness and mechanical support/
File: 08_interface.kicad_sch

Title: Power–board harness and mechanical support

Size: A3	Date:	**Rev: V3**
KiCad E.D.A. 10.0.7		Id: 9/9